

Response to the Theory Commission

On f , the Lattice Energy Expression, and What the Kish Action Principle Does and Does Not Determine

Prepared for KishLattice 16π Initiative LLC

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Standing of this response

Caveat

The commission was drafted by Atlas. This response is produced by the same underlying model. It brings time and rigour to the derivations that the commission could not, but it does *not* bring the independence a different reasoning system would. Where this response confirms arithmetic already in the commission's §1.3, that confirmation should be weighted as a re-check by the same hand, not as corroboration. Where it reaches a conclusion the commission did not anticipate — Q1 and Q4 below — those are the parts that carry information.

All arithmetic below was executed, not recalled. Every number can be reproduced from the stated inputs.

Summary of findings

Question	Finding
Q1 Does an action principle fix n ?	Not as written. Volume 1 contains no action functional. But one additional postulate — that the lattice bulk has four <i>spatial</i> dimensions — derives $n = 0$ exactly, by Gauss’s law. The postulate is independently motivated by the framework’s own 24-cell geometry. Its cost is stated.
Q2 Is ℓ_{cell} independently determined?	No, and it cannot be. Every input to the geometry is dimensionless. A length cannot be derived from dimensionless inputs. $\ell_{\text{cell}} = \ell_P$ is a postulate, not a result.
Q3 Which reading of f survives?	Under Q1, $f = \ell_{\text{cell}}^2 c^2$ is derived. The drag factor $F(d, T)$ is phenomenological and multiplies it; the referee’s reconciliation is consistent but only half-derived.
Q4 Is k_{geo} ’s dimensional status consistent?	No. Eq. 1.3 is correct: k_{geo} is dimensionless. §1.4 is in error. The Hubble relation $H_{\text{local}} = H_{\text{early}} + k_{\text{geo}}$ holds only in km/s/Mpc and fails by 18 orders of magnitude in SI. It is Erratum 4f in the theory.
Q5 Does a scale-bridging claim survive?	Not a dimensional one. $\pi/16$ fits the MOND $a_0 \sim cH_0$ coincidence no better than the literature’s $1/2\pi$. The framework’s only scale-free result is the dimensionless register structure, currently standing on one domain.

1 Q1 — The radial power

1.1 What Volume 1 actually contains

The commission asks whether the Kish Action Principle of Volume 1 §1.2 yields a determinate power n in

$$f = \ell_{\text{cell}}^2 c^2 \left(\frac{\ell_{\text{cell}}}{r} \right)^n. \quad (1)$$

Reading §1.2 in full: §1.2 defines $N_{\text{total}} = d^2 = 16$. §1.2.1 postulates $\Phi_{\text{cycle}} = \pi$. §1.2.2, titled “The Kish Action Principle,” defines $k_{\text{geo}} = N_{\text{total}}/\Phi_{\text{cycle}}$. §1.3 gives the Gutzwiller/Berry–Keating phase $\Phi_{\text{Kish}} = \cos[E \cdot \frac{16}{\pi} \cdot \ln p]$.

Negative result

There is no action functional anywhere in Volume 1. No $S = \int L dt$, no Lagrangian density, no variation, no Euler–Lagrange equation. The section titled “Action Principle” names a ratio. It does not state a principle of stationary action.

The Gutzwiller phase of §1.3 is a trace-formula ansatz, not a Lagrangian; it describes a spectral density, not the dynamics of a mass at radius r .

Consequently the question as posed has a negative answer: the framework as written does not determine n .

1.2 The one route that does

A negative answer is acceptable, but there is a constructive one available, and it uses only material the framework already contains. It requires making explicit a postulate that Volume 1 states in passing and never uses.

Derivation

Postulate (already in Volume 1, §1.2.2): “mapping a 4D hypercube onto a cyclic time loop.” The 24-cell of the Rosetta Atlas is a regular polytope in **four spatial dimensions**. Take the lattice bulk to be four-dimensional in space, with time the separate cyclic coordinate $\Phi = \pi$.

Gauss’s law in d spatial dimensions. For a point source of strength Q , the flux through a $(d - 1)$ -sphere of radius r is conserved:

$$\oint_{S^{d-1}} \mathbf{E} \cdot d\mathbf{A} = Q \implies E(r) \Omega_{d-1} r^{d-1} = Q \implies E(r) = \frac{Q}{\Omega_{d-1} r^{d-1}},$$

where Ω_{d-1} is the surface area of the unit $(d - 1)$ -sphere. Integrating the field gives the potential,

$$V(r) = -\int E dr = \frac{Q}{(d-2) \Omega_{d-1}} \cdot \frac{1}{r^{d-2}} \quad (d \neq 2). \quad (2)$$

For $d = 4$:

$$\Omega_3 = 2\pi^2, \quad V(r) = \frac{Q}{2 \cdot 2\pi^2} \cdot \frac{1}{r^2} = \frac{Q}{4\pi^2 r^2}.$$

The potential of a point source in four spatial dimensions falls as $1/r^2$. That is $n = 0$ in Equation (1), derived rather than assumed.

Interaction energy of a test mass m . With the source strength set by the lattice scale, $Q = \kappa m \ell_{\text{cell}}^2 c^2$ for some dimensionless coupling κ ,

$$E_{\text{lat}} = m V(r) = \frac{\kappa}{4\pi^2} m c^2 \left(\frac{\ell_{\text{cell}}}{r} \right)^2.$$

Matching to the Atlas form $E_{\text{lat}} = \frac{16}{\pi} m c^2 (\ell_{\text{cell}}/r)^2$ requires

$$\frac{\kappa}{4\pi^2} = \frac{16}{\pi} \implies \kappa = 64\pi.$$

Q1 — Answer

The Kish Action Principle as written does not fix n . Adding one postulate — *the lattice bulk is four-dimensional in space* — fixes $n = 0$ exactly, by Gauss’s law, and reproduces the Atlas form with a dimensionless coupling $\kappa = 64\pi$.

Is the postulate independently motivated? Yes, and more strongly than most of the framework’s other assumptions: the 24-cell is a four-dimensional object; Volume 1 explicitly invokes a 4D hypercube; and $N_{\text{total}} = d^2 = 16$ already uses $d = 4$. The postulate makes explicit what the geometry already assumes.

Its cost is severe and must be stated. We observe three spatial dimensions. Observed gravity and electrostatics both go as $1/r$, not $1/r^2$. **The $1/r^2$ lattice energy therefore cannot be the interaction we measure.** It can only be a bulk quantity whose *projection* onto the three-dimensional boundary recovers $1/r$ — and that projection is not derived anywhere. Volume 1’s holographic language (§1.2: “a holographic projection on the boundary preserves the full information content of the bulk tensor”) gestures at it. It does not compute it.

The framework has therefore traded one gap for a smaller one. Before: an undefined f and an unfixed power. After: a derived $f = \ell_{\text{cell}}^2 c^2$, a derived $n = 0$, a derived coupling $\kappa = 64\pi$, and one open problem — **the bulk-to-boundary projection** that would turn a 4D $1/r^2$ into the 3D $1/r$ we observe. That is a well-posed question. It is also the question on which the framework’s connection to observed physics now entirely rests.

1.3 What $\kappa = 64\pi$ would have to mean

The coupling absorbs the $16/\pi$ and the $4\pi^2$ of the 3-sphere. Written out:

$$\kappa = 64\pi = 4 \times 16 \times \pi = 4 N_{\text{total}} \Phi_{\text{cycle}}.$$

Four times the metric component count times the cyclic phase. Whether that factorisation is meaningful or numerological I cannot say from the material available. It is recorded so it can be tested against the geometry by someone who knows the 24-cell better than I do.

2 Q2 — Is ℓ_{cell} independently determined?

Negative result

No, and it cannot be determined from the framework’s internal geometry.

Every input to the derivation of k_{geo} is dimensionless: $N_{\text{total}} = 16$ is a count, $\Phi_{\text{cycle}} = \pi$ is an angle in radians, the 24-cell packing is a combinatorial fact. **No combination of dimensionless quantities yields a length.** The framework’s geometry, however rich, is scale-free by construction and cannot supply a scale.

$\ell_{\text{cell}} = \ell_P = \sqrt{\hbar G/c^3}$ is therefore a **postulate** — the identification of the lattice spacing with the Planck length — not a derivation. It imports $\hbar$, G and c from outside the framework. Volume 1’s Listing 14.1 asserts it as a variable assignment and does not argue it.

This is not a defect peculiar to KishLattice. *Every* discrete-spacetime programme faces it. It is simply a fact that should be stated as a postulate rather than presented as a consequence.

Q2 — Answer

ℓ_{cell} is fixed by postulate at $\ell_P = 1.616255 \times 10^{-35}$ m. The Rosetta Atlas’s alternative chain via $c = \ell_{\text{cell}} f_{24}$ with a laboratory f_{24} is unit-dependent and should be withdrawn; Volume 1’s identification is the one that survives.

With $\ell_{\text{cell}} = \ell_P$, the equipartition radius is

$$r^* = \frac{4}{\sqrt{\pi}} \ell_P = 2.256758 \times 1.616255 \times 10^{-35} \text{ m} = 3.648 \times 10^{-35} \text{ m},$$

2.26 Planck lengths. **There is no physical system at that radius.** The equipartition condition $E_{\text{lat}} = mc^2$ is therefore satisfied only in the Planck regime, which is consistent with the 4D-bulk reading of Q1 — the lattice energy is Planck-scale bulk physics, not an atomic or astrophysical quantity.

On the commission’s check of $4/\sqrt{\pi}$ against the electron length scales: confirmed. They are separated by $1/\alpha$, not by 2.2568. No coincidence exists, and none should be manufactured.

3 Q3 — Which reading of f survives?

Q3 — Answer

Under the Q1 derivation, reading **(b)** is correct and reading **(a)** is a separate multiplicative factor:

$$f = \underbrace{\ell_{\text{cell}}^2 c^2}_{\text{derived (Q1)}} \times \underbrace{F(d, T)}_{\text{phenomenological}}, \quad F(d, T) = \left[1 + k d \left(\frac{T_0}{T} \right)^\alpha \right]^{-1}.$$

The referee’s reconciliation is consistent, and exactly half of it is derived. The dimensional carrier $\ell_{\text{cell}}^2 c^2$ follows from Gauss’s law in four dimensions with the Planck identification. The drag factor $F(d, T)$ does not follow from anything in Volume 1 or the Atlas; it is a phenomenological damping law with three free constants (k, T_0, α) that must be fitted or independently derived.

Reading **(c)** — f as a stand-in for G — is dissolved by Q1. The $1/r^2$ form is not gravity in disguise; it is a four-dimensional point-source potential, and k_{geo} does not replace G because the bulk expression has no G in it. G enters only through $\ell_{\text{cell}} = \ell_P$.

The name “slipstream factor” should attach to $F(d, T)$ alone, which is what it describes. The full f of the Atlas conflates a derived geometric carrier with an undetermined damping law, and the conflation is the source of the three incompatible readings.

4 Q4 — The dimensional status of k_{geo}

Derivation

Equation 1.3 is correct. $k_{\text{geo}} = N_{\text{total}}/\Phi_{\text{cycle}} = 16/\pi$ is a count divided by an angle. Both are dimensionless. k_{geo} is a pure number.

§1.4 is in error. It asserts that $16/\pi$ possesses “the physical dimensions of Metric Tension per Unit Volume,” $[M][L^{-1}][T^{-2}]$, and writes

$$H_{\text{local}} = H_{\text{early}} + k_{\text{geo}}. \quad (3)$$

Test in the units Volume 1 uses. Planck 2018 gives $H_0 = 67.4 \text{ km/s/Mpc}$; SH0ES gives 73.0 ± 1.0 . Then

$$67.4 + 5.093 = 72.49 \quad \text{vs} \quad 73.0 \pm 1.0.$$

Inside the error bar. This is why §1.4 makes the claim.

Test in SI. $1 \text{ km/s/Mpc} = 3.2408 \times 10^{-20} \text{ s}^{-1}$, so

$$H_{\text{early}} = 2.1843 \times 10^{-18} \text{ s}^{-1}, \quad H_{\text{early}} + k_{\text{geo}} = 5.0930 \text{ s}^{-1},$$

which exceeds $H_{\text{SH0ES}} = 2.3658 \times 10^{-18} \text{ s}^{-1}$ by a factor of 2.15×10^{18} .

Negative result

Equation (3) holds in one unit system and fails by eighteen orders of magnitude in another. A physical relation cannot do that. A dimensionless number cannot be added to a rate. The agreement in km/s/Mpc is a numerical coincidence contingent on the choice of kilometre, second, and megaparsec.

This is Erratum 4f — unit dependence — appearing in the theory. The empirical programme found the same defect in its statistic and spent a volume removing it. The theory carries it in §1.4.

Q4 — Answer

There is no formulation in which both §1.3 and §1.4 are true. k_{geo} is dimensionless. §1.4's dimensional assertion is wrong, and the Hubble relation built on it is a unit coincidence and should be withdrawn.

What could be rescued. If the framework wishes to relate k_{geo} to the Hubble tension, the dimensionally admissible form is multiplicative:

$$H_{\text{local}} = H_{\text{early}} (1 + \epsilon), \quad \epsilon = \frac{73.0 - 67.4}{67.4} = 0.083 \pm 0.016.$$

Whether $\epsilon = 0.083$ relates to k_{geo} in any derivable way — $\pi/16$ something, $1/(4k_{\text{geo}}) = 0.049$, $\ln(k_{\text{geo}})/20 = 0.081$ — I checked the obvious candidates and none is compelling. The last is within 1σ and is exactly the kind of coincidence the commission's conditions say to distrust. I report it so it is not rediscovered as a result.

5 Q5 — Does a scale-bridging claim survive?

5.1 The MOND check

Milgrom noted early that MOND's acceleration scale is numerically close to cH_0 : $a_0 \approx cH_0/2\pi$. If the lattice supplied a bridge, k_{geo} might appear in that ratio.

Derivation

$$cH_0 = 2.99792 \times 10^8 \times 2.2685 \times 10^{-18} = 6.801 \times 10^{-10} \text{ m/s}^2 \quad (H_0 = 70 \text{ km/s/Mpc}),$$

$$a_0 = 1.20 \times 10^{-10} \text{ m/s}^2, \quad \frac{a_0}{cH_0} = 0.1764.$$

candidate	value	deviation
$1/2\pi$ (Milgrom)	0.1592	10%
$\pi/16 = 1/k_{\text{geo}}$	0.1963	11%
$1/6$	0.1667	6%

Negative result

$\pi/16$ fits the a_0/cH_0 ratio no better than the literature's $1/2\pi$, and worse than $1/6$. **There is no evidence that k_{geo} appears in the MOND scale.** The commission's caution about coincidences applies: with H_0 uncertain at the 8% level and a_0 at the 10–15% level, any constant between 0.15 and 0.20 “fits.”

5.2 The structural point

Q5 — Answer

A dimensional bridge does not survive, and the reason is structural.

Under Q1 and Q2, the lattice energy is a Planck-scale bulk quantity: $\ell_{\text{cell}} = \ell_P$, $r^* = 2.26 \ell_P$, and E_{lat} is negligible at every scale above the Planck regime ($4.75 \times 10^{-49} mc^2$ at the Bohr radius). **It has nothing to say at atomic scales and nothing to say at galactic scales.** It cannot bridge them because it is present at neither.

The commission's framing is correct: a characteristic length picks a scale, and picking a scale is the opposite of bridging scales. MOND uses an acceleration threshold precisely because a threshold is scale-free in a way a length is not.

The framework's one genuinely scale-free object is the register structure. Periodicity in $\log x$ has no preferred scale by construction — Volume 12 proved unit invariance for the period statistic by algebra — and a register at Δ_N means the same thing for pulsar spin-down at 10^{-21} s/s and NIST transitions at 10^{16} Hz. *If* the lattice bridges scales, that is where the claim lives: in a dimensionless log-period, not in a cell spacing.

That claim currently rests on one domain, and that domain faces a Rydberg null that has not been run. The empirical programme has been honest about this. The theory should be equally honest that E_{lat} is not the vehicle.

6 Errata implied by this response

Erratum implied

1. **Volume 1, §1.4.** The assertion that k_{geo} carries dimensions of metric tension per unit volume is withdrawn; k_{geo} is dimensionless per Eq. 1.3. The relation $H_{\text{local}} = H_{\text{early}} + k_{\text{geo}}$ is withdrawn as a unit coincidence.
2. **Volume 1, Listing 14.1.** The script divides $\ell_P f_P = c$ by k_{geo} and reports the result as c ; it is off by exactly $1/k_{\text{geo}}$. The “5-SIGMA ALIGNMENT LOCKED” status is a print statement, not a test. The listing is withdrawn. Separately, $c = \ell_P f_P$ is an identity of Planck units, not a derivation.
3. **Rosetta Atlas, Lattice Energy Equation.** f is redefined as $f = \ell_{\text{cell}}^2 c^2 \cdot F(d, T)$, with the geometric carrier derived (this response, Q1) and the drag factor labelled phenomenological. The parameter block states this.
4. **Rosetta Atlas, “mirrors Newton”.** The paragraph is struck. The expression is a four-dimensional point-source potential, not a gravitational analogue, and k_{geo} does not replace G .
5. **Rosetta Atlas, ℓ_{cell} via f_{24} .** The chain $c = \ell_{\text{cell}} f_{24}$ with $f_{24} = 24/\pi$ Hz is unit-dependent and contradicts Volume 1 by 42 orders of magnitude. Withdrawn in favour of $\ell_{\text{cell}} = \ell_P$, stated as a postulate.
6. **New postulate to be stated explicitly.** The lattice bulk has four spatial dimensions. This is already implicit in $N_{\text{total}} = d^2 = 16$ and in the 24-cell, and it is what derives $n = 0$. It should be a numbered postulate, not a passing phrase.
7. **New open problem to be stated explicitly.** The bulk-to-boundary projection that maps the 4D $1/r^2$ potential onto the observed 3D $1/r$. Nothing in the corpus computes it. Everything connecting E_{lat} to measured physics depends on it.

7 Closing

The commission asked whether a derivation exists. The answer is: **not in the documents as written, but one is available at the cost of one postulate the framework already half-states**, and it derives f , n , and a coupling constant in three lines. The price is that the lattice energy becomes a four-dimensional Planck-scale bulk quantity with no computed connection to the three-dimensional physics we observe.

That is not a breakthrough. It is a clarification. It replaces an undefined parameter with a derived one, and replaces a vague gap with a precise one — the projection problem — that a competent theorist could attack. Whether the framework survives that attack I cannot say.

The commission also hoped this response might open a bridge between the quantum and MOND regimes. It does not, and I think the reason is instructive: the framework’s dimensional machinery lives at the Planck scale, and its dimensionless machinery — the register structure — is the only part with scale-free content. If a bridge exists in this programme, it is in Volume 12’s log-periodicity result, not in Volume 1’s energy expression. The empirical programme has one domain standing and a Rydberg null to run. That is where I would put the effort.

Three of the five questions returned negative results. Under the commission's own conditions those are publishable, and I would publish them.